2.1.6 Composite systems

Very often one is interested in describing physical systems consisting of several distinct parts. In quantum mechanics, the mathematical structure which allows us to combine state spaces of multiple systems is the tensor product. Given two quantum systems A and B with state spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively, the state space of the joint system AB is $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. This is referred to as the *tensor product postulate*.

A basis for the joint state space can be constructed from the tensor products of basis states for each subsystem. If the sets $\{|n\rangle_A\}_n$ and $\{|m\rangle_B\}_m$ denote bases for $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively, then $\{|n\rangle_A \otimes |m\rangle_B\}_{n,m}$ is a basis for $\mathcal{H}_{AB}$ and any state of the joint system can be written

$$|\psi\rangle_{AB} = \sum_{n,m} \alpha_{nm} |n\rangle_A \otimes |m\rangle_B. \tag{2.32}$$

Since it can quickly become tedious and cluttered to write out all tensor products explicitly in longer derivations or for systems with multiple parts, many short-hand notations are used. For example, the state $|0\rangle_A \otimes |1\rangle_B$ might be denoted $|0\rangle_A|1\rangle_B$ or $|0,1\rangle_{AB}$ or just $|01\rangle_{AB}$, and often the subscript is also dropped, $|01\rangle$, when it is clear which is the first and second system etc. In these notes, we will use whichever notation is convenient in the given context.

2.2 Qubits

The smallest, interesting quantum system is a two-dimensional system, with state space corresponding to the smallest, non-trivial Hilbert space $\mathcal{H} = \mathbb{C}^2$. Such a two-dimensional quantum system is known as a 'quantum bit' or *qubit*. It represents the smallest unit of quantum information, in analogy with the smallest unit of classical information, the *bit*.

The classical bit can take two values, commonly denoted 0 and 1. Similarly, we can take $|0\rangle$ and $|1\rangle$ to denote two basis states of the qubit, commonly referred to as the 'computational basis'. The most general (pure) state of a qubit is

$$\alpha|0\rangle + \beta|1\rangle, \tag{2.33}$$

with $\alpha, \beta \in \mathcal{C}$. The state is normalised when $|\alpha|^2 + |\beta|^2 = 1$.

2.2.1 The Bloch sphere

Physically, any two-dimensional quantum system can be used to realise a qubit. For example, two energy levels of an atom, the polarisation of a photon or the spin states of a particle such as an electron. Mathematically, a qubit is equivalent to a spin-$\frac{1}{2}$ system. We may identify the states $|0\rangle$, $|1\rangle$ with the spin-up and -down states $|\uparrow\rangle, |\downarrow\rangle$ along a given direction, e.g. the z-axis. The coefficients in the general superposition (2.33) then define the orientation of the spin in three-dimensional space. Because a global phase has no physical significance, any normalised qubit state (2.33) can be written in terms of two real numbers as

$$\cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle. \tag{2.34}$$

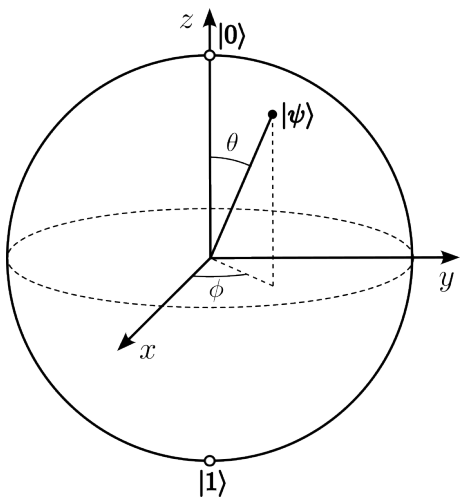

Figure 2.1: The Bloch sphere. Any qubit state on the form (2.34) corresponds to a point on the surface of the unit sphere, parametrised by the polar angle $\theta \in [0, \pi]$ and the azimuthal angle ϕ. The states at the extremal points along each axis x, y, z are the eigenstates of $\hat{\sigma}_x$, $\hat{\sigma}_y$, $\hat{\sigma}_z$. E.g. the states at the 'north and south poles' along z are $|0\rangle$ and $|1\rangle$. Points in the interior correspond to mixed states via (2.75). The origin corresponds to the maximally mixed state.

Here, θ and ϕ can be understood as the spherical coordinates of a point on the surface of the unit sphere in $\mathbb{R}^3$. This provides an intuitive and extremely useful visual representation of qubit states in terms of points on this unit sphere, known as the *Bloch sphere*. It is illustrated in Fig. 2.1. The vector pointing to a given state is known as the *Bloch vector* and can be thought of as a visual representation of the spin of the qubit.

2.2.2 Pauli observables and operators

A set of very important quantities in qubit systems are the Pauli operators or Pauli observables $\hat{\sigma}_x$, $\hat{\sigma}_y$, and $\hat{\sigma}_z$. They correspond to the three spatial components of angular momentum for spin-$\frac{1}{2}$ systems. The Pauli operators are both unitary and Hermitian

$$\hat{\sigma}_k^\dagger \hat{\sigma}_k = \hat{\sigma}_k \hat{\sigma}_k^\dagger = \mathbb{1}, \tag{2.35}$$

$$\hat{\sigma}_k^\dagger = \hat{\sigma}_k, \tag{2.36}$$

for $k = x, y, z$. This means they can be understood both as unitary operations that transform qubit states and as observables that can be measured. It also implies that each Pauli operator is its own inverse, $\hat{\sigma}_k^{-1} = \hat{\sigma}_k$ and hence $\hat{\sigma}_k^2 = \mathbb{1}$ for $k = x, y, z$. They obey the angular-momentum commutation relations

$$[\hat{\sigma}_x, \hat{\sigma}_y] = 2i\hat{\sigma}_z, \tag{2.37}$$

$$[\hat{\sigma}_y, \hat{\sigma}_z] = 2i\hat{\sigma}_x, \tag{2.38}$$

$$[\hat{\sigma}_z, \hat{\sigma}_x] = 2i\hat{\sigma}_y, \tag{2.39}$$